

A quadratic lower bound for every permutation-fair die

Research proof: three independent full mathematical reviews completed

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Status. Three independent full mathematical reviews completed, with no gap identified; this is not formal verification. The proof combines two independently reviewed inputs with an all-orders Gaussian estimate.

Theorem 1. *There is an absolute $c > 0$ such that every individual die in a tie-free permutation-fair collection of n equiprobable finite dice has at least cn^2 faces. More generally, for any monotone mixed-moment tensor with m columns, its first and last row weights are at most C/m^2 . Consequently the smallest possible size of one distinguished die, with all other dice unrestricted, is $\Theta(n^2)$.*

The last assertion uses the independently reviewed $O(n^2)$ construction in `polynomial_exceptional_die.tex`. No claim about literature novelty or about the optimal total number of faces is made here. The lower bound needs only the weaker condition that the distinguished die wins every subset containing it with probability equal to the reciprocal of that subset's size: these are precisely the mixed-moment identities used below. Uniformity of every complete order is unnecessary for this lower bound.

Notation and reviewed inputs

Assume $p_j(q) \in [0, 1]$ are nondecreasing in q , the positive row weights w_q sum to one, and

$$\mathbb{E}_w \prod_{j \in S} p_j = 1/(|S| + 1) \quad (S \subseteq [m]).$$

Write

$$u_j = 1 - p_j, \quad s = \sum_j u_j, \quad \ell = \lfloor m/4 \rfloor, \quad N = m - \ell, \quad \alpha = \ell/m, \quad x = \alpha s, \quad \theta = 1/16.$$

For a uniform ℓ -subset B , put $A_B = \prod_{j \in B} p_j$, $\bar{A} = \mathbb{E}_B A_B$, and $v_B = N^{-1} \sum_{j \notin B} \ell u_j$. The probability measure μ on x has row masses $(\ell + 1)w_q \bar{A}(q)$. The measure ν has row masses mw_q and total mass m . Let β have probability density

$$b_\ell(z) = \frac{\ell + 1}{\ell} (1 - z/\ell)^\ell \mathbf{1}_{[0, \ell]}(z).$$

All constants below are absolute. We work with sufficiently large m ; enlarging constants covers the remaining range.

The two crucial reviewed inputs are

$$\max_j u_j \leq C(s + \delta)/m, \quad \delta = m^{-1/3}, \quad (1)$$

$$\nu([0, t]) \leq C(t + h) \quad (t \geq 0), \quad h = m^{-3/4}. \quad (2)$$

The first is Theorem 1 of `../rational_designs/asymptotic/quantitative_relative_endpoint_balance.tex`. The second is the elementary Gaussian bootstrap from exponent 1/2 to 3/4 in `elementary_endpoint_cdf_bootstrap.tex`. Only this single step is needed; no asymptotic Laguerre estimate or constants depending on a limiting exponent are used.

We also use the earlier exact polarization identity

$$\beta(P) = (\ell + 1) \mathbb{E}_w \mathbb{E}_B [A_B \text{Pol}_N(P)((\ell u_j)_{j \notin B})] \quad (3)$$

for $\deg P \leq N$, and the elementary estimate

$$\bar{A} \leq e^{-\alpha s} = e^{-x}. \quad (4)$$

For a centered real vector y of length N , with $V = \sum_j y_j^2$, the entire-function Cauchy estimate gives

$$|e_k(y)| / \binom{N}{k} \leq (\sqrt{ekV}/N)^k. \quad (5)$$

These facts and the Gaussian Laguerre identities are proved in `../power_per_die_lower_bound.tex` and `gaussian_laguerre_variance_lemmas.tex`.

An all-orders estimate for the particular endpoint tests

Set

$$T_d(y) = \sum_{j < d} (1-j/d) L_j(y) = d^{-1} L_{d-1}^{[2]}(y), \quad U_d(z) = d^{-2} T_d(\theta z)^2, \quad R_d(z) = (2/(\theta d) - z) T_d(\theta z)^2.$$

Square brackets denote an associated-Laguerre parameter.

Lemma 1. *For every $d \geq 1$, $0 \leq j \leq d-1$, and $z > 0$,*

$$\left| \frac{d^j}{dz^j} T_d(\theta z) \right| \leq \frac{C e^{\theta z/2}}{z} \left(C \sqrt{d/z} \right)^j. \quad (6)$$

Higher derivatives vanish. Consequently, for $z \geq 1/(2d)$ and every k ,

$$\frac{|R_d^{(k)}(z)|}{k!} \leq \frac{C(k+1)e^{\theta z}}{z} \frac{(C\sqrt{d/z})^k}{k!}, \quad (7)$$

$$\frac{|U_d^{(k)}(z)|}{k!} \leq \frac{C e^{\theta z}}{d^2 z^2} \frac{(C\sqrt{d/z})^k}{k!}. \quad (8)$$

All constants are uniform in the derivative order.

Proof. The off-diagonal Gaussian matrix-coefficient bound gives, for integers $r, a \geq 0$,

$$y^{a/2} |L_r^{[a]}(y)| \leq e^{y/2} \sqrt{(r+a)!/r!}.$$

Apply it with $r = d-j-1$, $a = j+2$, whose sum is always $d+1$. Since

$$D_z^j T_d(\theta z) = (-\theta)^j L_{d-j-1}^{[j+2]}(\theta z)/d,$$

and $(d+1)!/(d-j-1)! \leq (d+1)^{j+2}$, it proves (6). This is valid simultaneously for every derivative order, rather than only for fixed order. The product rule proves (8). For (7), use

$$R_d^{(k)} = (2/(\theta d) - z) (T_d(\theta z)^2)^{(k)} - k (T_d(\theta z)^2)^{(k-1)}.$$

When $z \geq 1/(2d)$, the linear factor has magnitude at most Cz , and the second term costs at most an additional factor $k/\sqrt{dz} \leq \sqrt{2}k$. This proves the assertion. $\square$

We will sum the whole polarization expansion, using

$$\sum_{k=2}^{\infty} (k+1)a^k k^{k/2}/k! \leq Ca^2 e^{Ca^2} \quad (a \geq 0). \quad (9)$$

For completeness, split into even and odd orders. At $k = 2j$, $(2j)^j/(2j)! \leq 2^j/j!$, since $(2j)!/j! \geq j^j$. At $k = 2j+1$, the same argument bounds the ratio by $C^{j+1}(j+1)/j!$. Summing the two exponential series gives $C(a^2 + a^3)(1 + a^4)e^{Ca^2}$; increasing C absorbs the extra factors and gives (9). This argument supplies absolute constants.

Improved subset moments and lower tails

On rows with $s \leq m/256$, the success-weighted subset law $\Pr(B) = A_B/e_\ell(p)$ has $\Delta = v_B - x$ satisfying

$$0 \leq \mathbb{E}\Delta \leq Cx(x+\delta)/m, \quad \mathbb{E}\Delta^2 \leq C\{x(x+\delta)/m + x^2(x+\delta)^2/m^2\}. \quad (10)$$

Indeed the exact bias identity and negative inclusion covariances proved in `../rational_designs/asymptotic/weighted_success_subset_moments.tex` bound the bias and variance by constants times $\sum_j u_j^2$. By (1), this sum is at most $Cs(s+\delta)/m$; also $p_j \geq 1/2$ here, and $x \asymp s$.

For the uniform subset law, the sharpened lower-tail bound is

$$\Pr_B(v_B < x/2) \leq \exp[-cmx/(x+\delta)]. \quad (11)$$

To verify it, the population $z_j = \ell u_j$ has mean x and maximum at most $C(s+\delta)$. For any upper bound R on that maximum, Maclaurin's inequality gives

$$\mathbb{E}_B e^{-Nv_B/R} \leq \left(m^{-1} \sum_j e^{-z_j/R} \right)^N \leq \exp[-Nx(1 - e^{-1})/R].$$

Markov's inequality gives (11), as in the earlier bootstrap. When success weighting is present, the unnormalized bad contribution is still bounded by (11): explicitly $\bar{A} \Pr_{\text{weighted}}(\text{bad}) = \mathbb{E}_{B \text{ uniform}} A_B \mathbf{1}_{\text{bad}} \leq \Pr_{\text{uniform}}(\text{bad})$.

Comparison at linear degree

Lemma 2. *There is an absolute $\kappa_0 > 0$ such that, for each fixed $0 < \kappa \leq \kappa_0$ and $d = \lfloor \kappa m \rfloor$,*

$$|\mu(R_d) - \beta(R_d)| \leq C\kappa + o(1), \quad (12)$$

$$|\mu(U_d) - \beta(U_d)| = o(1/m). \quad (13)$$

The constant C is independent of κ ; the little-oh terms may depend on the fixed positive κ .

Proof. Choose κ_0 small enough that $2d \leq N$ and $x \leq d$ implies $s \leq m/256$. We again compare exact polarization to $P(v_B)$ and then compare $P(v_B)$ to $P(x)$.

First consider $x \leq 1/d$. The global Gaussian bounds, with coefficient product $B_R = O(d)$ or $B_U = O(1)$ and variance proxy

$$\sum_{j \notin B} (z_j - v_B)^2 \leq CN(s+\delta)v_B,$$

give full polarization error, after subset averaging, at most $CB_P(d/m)(x + \delta)$ in this range. Indeed its series starts at order two and (9) has argument squared $O((d/m)(s + \delta)) = o(1)$. The weighted Taylor argument using (10) gives the same bound. One can use the global derivative envelopes $CB_P\sqrt{d/z}e^{\theta z}$ and $CB_P(d/z)e^{\theta z}$: the factor x in (10) cancels the singular Taylor remainder, as in the reviewed power-bound proof. At $x = 0$ all variables vanish and both errors are zero. By (2), the integrated small- x contributions are therefore

$$E_R^{\text{small}} \leq C\kappa d(1/d + \delta)(1/d + h) = o(1), \quad (14)$$

$$E_U^{\text{small}} \leq C\kappa(1/d + \delta)(1/d + h) = o(1/m). \quad (15)$$

Here $d \asymp m$, $\delta = m^{-1/3}$, $h = m^{-3/4}$, and hence $d\delta h \rightarrow 0$ and $\delta h = o(1/m)$.

Next let $1/d < x \leq d$. A subset is good when $v_B \geq x/2$; always $v_B \leq x/(1 - \alpha) < 2x$. Combining (7)–(8) with (5) cancels every power $v_B^{-k/2}$. On good subsets, (9) bounds the entire polarization error by

$$CW_P(x)e^{\theta v_B} \frac{d(s + \delta)}{m} \exp[Cd(s + \delta)/m], \quad W_R(x) = 1/x, \quad W_U(x) = 1/(d^2 x^2).$$

Multiply by A_B and average. Since $\bar{A} \leq e^{-x}$, $s \asymp x$, and $d/m \leq \kappa$, choosing κ_0 sufficiently small leaves a fixed exponential margin. Thus the respective unnormalized row errors are bounded by

$$C\kappa e^{-cx}(1 + \delta/x), \quad (16)$$

$$\frac{C}{dm} e^{-cx}(1/x + \delta/x^2). \quad (17)$$

These quantities are subsequently integrated against a constant multiple of ν , because $(\ell + 1)w \asymp mw$.

For bad subsets use the coarser global Gaussian bounds for the whole series. They are at most a polynomial in m times e^{Cx} in this region. Multiplication by $A_B \leq 1$ and (11) make the bad contribution negligible. If $1/d < x \leq \delta$, the exponent is at most $-cm/(d\delta)$, which is $-c/(\kappa\delta)$; if $x \geq \delta$, it is at most $-cm + Cd \leq -cm/2$ after reducing κ_0 . Both are superpolynomially small. Integration against total mass m preserves this conclusion.

For the weighted averaging error, keep the full linear bias $P'(x)\mathbb{E}\Delta$ and split only the Taylor remainder into good and bad subsets. On good subsets all intermediate points are in $[x/2, 2x]$. For R_d , the first two derivative bounds are

$$|R'_d| \leq C\sqrt{d}x^{-3/2}e^{2\theta x}, \quad |R''_d| \leq Cdx^{-2}e^{2\theta x}.$$

For U_d they are

$$|U'_d| \leq Cd^{-3/2}x^{-5/2}e^{2\theta x}, \quad |U''_d| \leq Cd^{-1}x^{-3}e^{2\theta x}.$$

With (10), and after multiplication by $\bar{A}$, these give the same envelopes (16)–(17). For example, the leading quadratic terms are exactly of these forms; the ratio of the linear term to the corresponding envelope is at most $C\sqrt{x/d} \leq C$. The squared-bias term introduces only $x(x + \delta)/m$, whose remaining polynomial factor is absorbed in the exponential margin. On bad subsets bound the remainder by $|P(v_B)| + |P(x)| + |P'(x)||\Delta|$. This is polynomial in m times e^{Cx} ; the weighted normalization cancels exactly as explained after (11). The same negligible bad probability applies.

For $x > d$, use the coarser global Gaussian full-series bound, with coefficient product B_P . Equations (9) and (4) leave e^{-cx} once κ_0 is small. Its integral is bounded by a polynomial in m times e^{-cd} . For averaging, the nonsingular first derivative bound and uniform-subset Cauchy–Schwarz give the same conclusion. These contributions are superpolynomially small because $d \asymp m$.

It remains to integrate (16)–(17). Stieltjes integration by parts under (2) gives

$$\begin{aligned} \int_{1/d}^{\infty} e^{-cx} d\nu(x) &\leq C, \\ I_1 &:= \int_{1/d}^{\infty} e^{-cx} x^{-1} d\nu(x) \leq C(\log d + hd), \\ I_2 &:= \int_{1/d}^{\infty} e^{-cx} x^{-2} d\nu(x) \leq C(d + hd^2). \end{aligned}$$

One may bound the cutoff endpoint term by $\nu([0, 1/d])$ times the endpoint value; the same displayed bounds result, including possible atoms. Hence the middle R error is

$$C\kappa(1 + \delta I_1) \leq C\kappa[1 + \delta \log d + \delta hd] = C\kappa + o(1).$$

The middle U error is at most

$$\frac{C}{dm}(I_1 + \delta I_2) \leq C \left[\frac{\log d}{dm} + \frac{h}{m} + \frac{\delta}{m} + \frac{\delta hd}{m} \right] = o(1/m).$$

Together with (14)–(15), the negligible tails, and (3), this proves the lemma. $\square$

The endpoint and its atom

The scalar taper calculation in the reviewed power-bound proof gives

$$\beta(R_d) \geq 1/(96\theta) > 0.$$

First choose a fixed $\kappa > 0$ small enough that the $C\kappa$ in (12) is less than half this constant, and then let m be sufficiently large. Since $R_d(z) \leq 0$ for $z \geq 2/(\theta d)$, positivity of $\mu(R_d)$ forces some row below that threshold. The last row minimizes x , so

$$x_K < 2/(\theta d).$$

Let $r = \lfloor d/16 \rfloor$. Then $r \asymp m$ and the comparison lemma applies with the smaller fixed proportionality constant. For $j < r$, $j\theta x_K \leq 1/8$, so the elementary estimate $|L_j(\theta x_K) - 1| \leq e^{j\theta x_K} - 1 < 1/3$ gives $U_r(x_K) \geq c > 0$. On the other hand orthogonality and $b_\ell(z) \leq 2e^{-\theta z}$ give $\beta(U_r) \leq C/r$. Therefore

$$c\mu(K) \leq \mu(U_r) \leq C/r + o(1/m) \leq C/m.$$

Finally $s_K = x_K/\alpha = O(1/m)$ and $A_B(K) \geq 1 - s_K \geq 1/2$. Thus

$$(\ell + 1)w_K/2 \leq \mu(K) \leq C/m, \quad w_K \leq C/m^2.$$

Reversal of rows and complementation of all columns preserve the mixed moments and give the same bound for the first row. For a distinguished equiprobable die, every row weight is $1/K$ and $m = n - 1$, proving Theorem 1.